13. Register Write Protection Function

The register write protection function protects important registers from being overwritten for in case a program runs out of control. The registers to be protected are set with the protect register (PRCR).

Table 13.1 lists the association between the PRCR bits and the registers to be protected.

Table 13.1 Association between PRCR Bits and Registers to be Protected

PRCR Bit	Register to be Protected
PRC0	• Registers related to the clock generation circuit: SCKCR, SCKCR2, SCKCR3, PACKCR, PLLCR, PLLCR2, PPLLCR, PPLLCR2, PPLLCR3, BCKCR, MOSCCR, SOSCCR, LOCOCR, ILOCOCR, HOCOCR, HOCOCR2, OSTDCR, OSTDSR, CKOCR
PRC1	• Registers related to the operating modes: SYSCR0, SYSCR1 • Registers related to the low power consumption functions: SBYCR, MSTPCRA, MSTPCRB, MSTPCRC, MSTPCRD, OPCCR, RSTCKCR, DPSBYCR, DPSIER0 to DPSIER3, DPSIFR0 to DPSIFR3, DPSIEGR0 to DPSIEGR3 • Registers related to clock generation circuit: MOSCWTCR, SOSCWTCR, MOFCR, HOCOPCR • Software reset register: SWRR
PRC3	• Registers related to the LVD: LVCMPCR, LVDLVLR, LVD1CR0, LVD1CR1, LVD1SR, LVD2CR0, LVD2CR1, LVD2SR

13.1 Register Descriptions

13.1.1 Protect Register (PRCR)

Address(es): 0008 03FEh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	PRKEY[7:0]								—	—	—	—	PRC3	—	PRC1	PRC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Function	R/W
b0	PRC0	Protect Bit 0	Enables writing to the registers related to the clock generation circuit. 0: Write disabled 1: Write enabled	R/W
b1	PRC1	Protect Bit 1	Enables writing to the registers related to operating modes, clock generation circuit, low power consumption, and software reset. 0: Write disabled 1: Write enabled	R/W
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b3	PRC3	Protect Bit 3	Enables writing to the registers related to the LVD. 0: Write disabled 1: Write enabled	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b8	PRKEY[7:0]	PRC Key Code	These bits control permission and prohibition of writing to the PRCR register. To modify the PRCR register, write A5h to the eight higher-order bits and the desired value to the eight lower-order bits as a 16-bit unit.	R/(W)*1

Note 1. Written values are not retained. These bits are read as 00h.

PRCi Bits (Protect Bit i) (i = 0, 1, 3)

These bits enable or disable writing to the corresponding registers to be protected.

Setting the PRCi bits to 1 and 0 enables and disables writing to the corresponding registers to be protected, respectively.